SEC P vs NP log 2026-05-10

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-17 19:17 UTC

SEC P vs NP — daily log 2026-05-10

61 cycles recorded

Tropical Convex Hull Extreme Points Lower Bound for ACC^0 Circuit Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Tropical Convex Geometry × ACC^0 Circuit Size
- **Recorded**: 2026-05-10 00:08 UTC
- **Entry ID**: `0bb43dada997`

Statement

For any CNF formula Φ with n variables, the number of extreme points in the tropical convex hull of its clause-indicator polynomial is Ω(log n) if Φ has no ACC^0 circuit of size n^c for any constant c > 0.

Rationale

Tropical convex hulls encode clause interactions via min-plus geometry, and their extreme points reflect combinatorial constraints on circuit representations. ACC^0 circuits' limited depth imposes structural bounds on these hulls, creating a bridge between algebraic geometry and circuit complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c44b66b7.py", line 78, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c44b66b7.py", line 56, in run_trial
    poly = clause_indicator_polynomial(clauses, n)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c44b66b7.py", line 42, in clause_indica
    poly = [0] * (2 ** n)
           ~~~~^~~~~~~~~~
```

```
MemoryError
```

Judge reasoning

Test crashed with MemoryError before producing results, preventing evaluation of conjecture support/falsification. | next: Optimize clause_indicator_polynomial to handle larger n via sparse representations or incremental construction

---

Kronecker Coefficient Exponential Gap in Symmetric Powers of Permanent vs Determinant

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Representation Theory of Symmetric Groups × Communication Complexity of Disjointness
- **Recorded**: 2026-05-10 00:22 UTC
- **Entry ID**: `722bbadc85c5`

Statement

For all n ≤ 40, the Kronecker coefficient for the decomposition of $\mathrm{Sym}^{k(\mathrm{Sym}}n(V))$ into irreducible representations is exponentially larger for the permanent than for the determinant when $k = \lceil n/2 \rceil$, implying a $\Omega(n)$ communication complexity lower bound for the corresponding disjointness instance.

Rationale

Kronecker coefficients capture the structure of symmetric power decompositions, which are central to GCT's orbit closure separations. Linking them to communication complexity provides a bridge between algebraic geometry and combinatorial lower bounds, leveraging the exponential sensitivity of symmetric tensors to combinatorial constraints.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
nterexample': 'Ratio 1.0 is not exponentially larger than 2^7'}
TRIAL: {'metric_name': 'Kronecker Coefficient Ratio', 'metric_value': 1.0, 'instances_tested': 1, 'co
TRIAL: {'metric_name': 'Kronecker Coefficient Ratio', 'metric_value': 1.0, 'instances_tested': 1, 'co
TRIAL: {'metric_name': 'Kronecker Coefficient Ratio', 'metric_value': 1.0, 'instances_tested': 1, 'co
TRIAL: {'metric_name': 'Kronecker Coefficient Ratio', 'metric_value': 1.0, 'instances_tested': 1, 'co
TRIAL: {'metric_name': 'Kronecker Coefficient Ratio', 'metric_value': 1.0, 'instances_tested': 1, 'co
TRIAL: {'metric_name': 'Kronecker Coefficient Ratio', 'metric_value': 1.0, 'instances_tested': 1, 'co
TRIAL: {'metric_name': 'Kronecker Coefficient Ratio', 'metric_value': 1.0, 'instances_tested': 1, 'co
TRIAL: {'metric_name': 'Kronecker Coefficient Ratio', 'metric_value': 1.0, 'instances_tested': 1, 'co
TRIAL: {'metric_name': 'Kronecker Coefficient Ratio', 'metric_value': 1.0, 'instances_tested': 1, 'co

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_4e40f8cb.py", line 81, in <module>
    first_failing_seed = next(r["seed"] for r in res
```

Judge reasoning

Test crashed before completing evaluation, but partial results show counterexamples where ratio 1.0 fails to meet exponential thresholds. | next: Fix seed handling in test code and re-run with full coverage of $n \leq 40$ cases

---

Operator Norm Lower Bound for Disjointness Communication Matrices

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Operator Spaces Theory × Communication Complexity of DISJOINTNESS
- **Recorded**: 2026-05-10 00:30 UTC
- **Entry ID**: `4da9869ae44e`

Statement

For any n×n matrix M representing the DISJOINTNESS function, the completely bounded norm ||M||_{cb} ≥ Ω(n). This holds for all matrices where M[i,j] = 1 iff the i-th and j-th subsets are disjoint, with random subset generation using uniform distribution over 2^{⌈log$_2$n⌉}-element sets.

Rationale

Operator spaces theory provides tools to analyze tensor norms that capture the structure of communication matrices. The cb-norm, being invariant under tensor products, could expose inherent dimensionality constraints in DISJOINTNESS that classical discrepancy methods miss.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dc72256d.py", line 78, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dc72256d.py", line 60, in run_trial
    cb_norm_max, _ = numerical_range(M)
                     ^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dc72256d.py", line 46, in numerical_ran
    B = [M[i] - (i+1)*I for i in range(n)]
         ~~~~~^~~~~~~~~
TypeError: unsupported operand type(s) for -: 'list' and 'list'
```

Judge reasoning

Test crashed due to TypeError in matrix operations, preventing result collection | next: Fix the numerical_range function's matrix subtraction logic to handle list operations properly

---

Tensor Rank Exponential Gap in Read-Twice vs Read-Once Branching Programs for IP_2

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Tensor Rank over Real Numbers × Read-Twice Branching Programs
- **Recorded**: 2026-05-10 00:53 UTC
- **Entry ID**: `ec8a0368610a`

Statement

For any read-once branching program P over n variables, the tensor rank of the transition tensor T(P) is O(log n). For the IP_2 function, any read-twice branching program Q must satisfy rank(T(Q)) ≥ 2^{n/2} / poly(n).

Rationale

Tensor rank captures the algebraic complexity of branching program transitions. Read-once BPs decompose as low-rank tensors, while read-twice BPs require high-rank tensors to model their repeated access to variables, mirroring the exponential gap in communication complexity for IP_2.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6ec1712e.py", line 135, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6ec1712e.py", line 90, in run_trial
    read_once_matrix = matrix_multiplication(*read_once_bp)
                       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
TypeError: matrix_multiplication() takes 2 positional arguments but 5 were given
```

Judge reasoning

Test crashed due to incorrect argument count in matrix multiplication, preventing result collection | next: Fix matrix_multiplication argument count in test_6ec1712e.py

---

Projective Plane Line Count Bounds ABP Width for Monotone Circuits

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Finite Geometry × Monotone Boolean Circuit Size
- **Recorded**: 2026-05-10 01:02 UTC
- **Entry ID**: `96cc863111fe`

Statement

For every monotone Boolean circuit computing the characteristic function of a projective plane PG(2, q), the minimal width of an ABP computing the same function satisfies width ≥ $q^2 + q + 1$. This bound is tight for q = 2^k.

Rationale

Projective planes exhibit highly symmetric incidence structures that could encode combinatorial constraints mirroring the layered computation of ABPs. The tightness of the bound suggests a deep algebraic relationship between geometric incidence and algebraic computation.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_55481e38.py", line 112, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_55481e38.py", line 80, in run_trial
    points, lines = generate_projective_plane(q)
                    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_55481e38.py", line 20, in generate_proj
    raise ValueError("q must be at least 2")
ValueError: q must be at least 2
```

Judge reasoning

Test crashed due to invalid q value (must be ≥2), preventing evaluation of conjecture | next: Run test with q=2 and valid projective plane parameters to verify bounds

---

Quantum Rank Lower Bounds for Disjointness Communication Matrices

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Noncommutative Geometry × Communication Complexity of DISJOINTNESS
- **Recorded**: 2026-05-10 01:08 UTC
- **Entry ID**: `8dbbdbbc746a`

Statement

For the n×n DISJOINTNESS matrix M_n, the noncommutative rank τ(M_n) satisfies τ(M_n) ≥ c·n for some universal constant c > 0. This invariant is defined as the minimal k such that M_n can be expressed as a sum of k rank-1 matrices in the matrix algebra over the noncommutative ring $\mathbb{C}\langle x,y\rangle/(x^2=x,\ y^2=y)$.

Rationale

Noncommutative geometry provides tools to analyze matrix factorizations through ring-theoretic structures. The DISJOINTNESS matrix's inherent combinatorial structure may enforce a nontrivial lower bound on its noncommutative rank, revealing algebraic obstructions to efficient communication protocols.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_82a6913d.py", line 91, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_82a6913d.py", line 67, in run_trial
    tau_M_n = noncommutative_rank(M_n, n)
              ^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_82a6913d.py", line 58, in noncommutativ
    A = [[x[i][j] - y[j] * M[i][n + j] for j in range(n)] for i in range(n)]
          ~~~~~~~~^~~~~~~~~~~~~~~~~~~~
TypeError: unsupported operand type(s) for -: 'int' and 'list'
```

Judge reasoning

Test crashed due to TypeError, preventing result collection. Pre-registered support condition cannot be evaluated without successful trials. | next: Debug the noncommutative_rank function's matrix construction to resolve type mismatches in the algebraic operations

---

Symmetric Tensor Rank Gap in Permanent vs Determinant Decompositions

- **Verdict**: INCONCLUSIVE
- **Bridge**: Representation Theory of Symmetric Groups × Monotone Circuit Size for 3-CNFs
- **Recorded**: 2026-05-10 01:23 UTC
- **Entry ID**: 8de14a1366b2

Statement

For a random 3-CNF formula Φ with n variables, the dimension of the irreducible constituent λ in the symmetric square of the permanent's representation (computed via Young diagrams) exceeds the dimension of the corresponding constituent in the determinant's representation by a factor of Ω(2^{n/4}) with probability ≥ 0.95 over Φ.

Rationale

Symmetric tensor decompositions reveal structural asymmetry between permanent and determinant. The Young diagram dimensions capture combinatorial complexity, and their exponential gap could imply inherent circuit size separation via representation-theoretic obstructions.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6dfe40e5.py", line 110, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
```

```
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6dfe40e5.py", line 83, in run_trial
    perm_val = permanent(n, perm_matrix)
               ^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6dfe40e5.py", line 42, in permanent
    submatrix = [[matrix[i][k] for k in range(j) + range(j+1, n)] for i in range(1, n)]
                                        ~~~~~~~~~^~~~~~~~~~~~~~~
TypeError: unsupported operand type(s) for +: 'range' and 'range'
```

Judge reasoning

Test crashed due to Python TypeError in range concatenation, preventing data collection. Support condition (80% success rate) cannot be evaluated without valid results. | next: Fix the range concatenation error in the permanent function by using list concatenation (e.g., list(range(j)) + list(range(j+1, n)))

Negative Eigenvalue Count Lower Bounds SOS Degree for Max-CUT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Real Algebraic Geometry × SOS Degree for Max-CUT
- **Recorded**: 2026-05-10 01:47 UTC
- **Entry ID**: `510fd15279df`

Statement

For any n-variable Max-CUT instance, the minimal SOS degree d required to certify its integrality gap satisfies d ≥ k + 1, where k is the number of negative eigenvalues of the moment matrix of the degree-d pseudoexpectation.

Rationale

The number of negative eigenvalues quantifies the ‘deficiency’ in positive semidefiniteness of the moment matrix, which directly correlates with the need for higher-degree SOS constraints to resolve non-positivity in the polynomial representation of the Max-CUT objective.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.14s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0ee3b164.py", line 116, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0ee3b164.py", line 91, in run_trial
    eigenvalues = [abs(e) for e in gaussian_elimination(M)[-1]]
                                   ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_0ee3b164.py", line 28, in gaussian_elim
    factor = 1 / A[i][i]
             ~~^~~~~~~~~
ZeroDivisionError: division by zero
```

Judge reasoning

Test crashed with ZeroDivisionError, preventing data collection. Pre-registered support condition cannot be evaluated without valid results. | next: Debug the gaussian_elimination function to handle singular matrices, then re-run tests with multiple seeds

---

Plethysm Coefficient Gap in Homogeneous Polynomial Decompositions

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Plethysm Theory × Boolean Circuit Size
- **Recorded**: 2026-05-10 03:39 UTC
- **Entry ID**: `cc61e4e0da28`

Statement

For any n ≥ 2, the plethysm coefficient of the permanent polynomial perm_n decomposed into symmetric squares of GL(n) representations exceeds that of det_m for all m < n^{1.5}, with ratio ≥ 2^{Ω(n)}

Rationale

Plethysm coefficients encode symmetry-breaking patterns in polynomial representations. If permanent's decomposition requires exponentially more symmetric squares than determinant's, this could expose structural asymmetries exploitable for circuit lower bounds.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.00s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing verification of support fraction or counterexamples. | next: Run test with extended timeout and higher resource allocation

---

Hypergraph Maximum Matching Inverse Proportional to ACC^0 Circuit Size for 3-CNFs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Hypergraph Theory × ACC^0 Circuit Size
- **Recorded**: 2026-05-10 03:58 UTC
- **Entry ID**: `9d4ad01564fe`

Statement

For any 3-CNF formula with n variables, the maximum matching size of its clause hypergraph is Θ(log n) if and only if the formula can be computed by an ACC^0 circuit of size Θ(n^k) for some constant k.

Rationale

The maximum matching size captures the sparsity of clause overlaps. Sparse overlaps (logarithmic) suggest a structured dependency that can be captured by ACC^0 circuits, which have limited depth but can handle structured patterns.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
'max_matching_size', 'metric_value': 85, 'instances_tested': 1, 'conjecture_holds': False, 'countere
TRIAL: {'metric_name': 'max_matching_size', 'metric_value': 23, 'instances_tested': 1, 'conjecture_h
TRIAL: {'metric_name': 'max_matching_size', 'metric_value': 153, 'instances_tested': 1, 'conjecture_
TRIAL: {'metric_name': 'max_matching_size', 'metric_value': 143, 'instances_tested': 1, 'conjecture_
TRIAL: {'metric_name': 'max_matching_size', 'metric_value': 38, 'instances_tested': 1, 'conjecture_h
TRIAL: {'metric_name': 'max_matching_size', 'metric_value': 239, 'instances_tested': 1, 'conjecture_
TRIAL: {'metric_name': 'max_matching_size', 'metric_value': 127, 'instances_tested': 1, 'conjecture_
TRIAL: {'metric_name': 'max_matching_size', 'metric_value': 255, 'instances_tested': 1, 'conjecture_
TRIAL: {'metric_name': 'max_matching_size', 'metric_value': 133, 'instances_tested': 1, 'conjecture_
TRIAL: {'metric_name': 'max_matching_size', 'metric_value': 124, 'instances_tested': 1, 'conjecture_

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_70a9fc8c.py", line 82, in <module>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_70a9fc8c.py", line 82, in <genexpr>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                              ~^^^^^^^^
KeyError: 'seed'
```

Judge reasoning

Test crashed before completing validation; counterexamples in output may be unreliable | next: Re-run test with increased time limits and deterministic seed configuration

---

Finite Field Rank and Branching Program Width for Boolean Functions

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Finite Field Linear Algebra × Branching Program Width
- **Recorded**: 2026-05-10 04:08 UTC
- **Entry ID**: `89d7e3ddf0de`

Statement

For a Boolean function f on n variables, let M_f be the matrix over GF(2) with rows representing variables and columns representing truth assignments, where M_f[i][j] = f(x_j) if x_j has i-th variable set. The minimal branching program width for f is at least the rank of M_f over GF(2).

Rationale

The rank of M_f captures the linear dependencies in f's truth table. A higher rank implies more complex interactions between variables, necessitating wider branching programs to represent the function's behavior. This link could expose algebraic structure in complexity measures.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
TRIAL: {'metric_name': 'rank vs. BDD width', 'metric_value': 10, 'instances_tested': 1, 'conjecture_h
TRIAL: {'metric_name': 'rank vs. BDD width', 'metric_value': 10, 'instances_tested': 1, 'conjecture_h

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b46512cb.py", line 72, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b46512cb.py", line 56, in run_trial
    bdd_width_value = bdd_width(M_f)
                      ^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b46512cb.py", line 45, in bdd_width
    if truth_table[i][var] != truth_table[2 * i][var]:
                              ~~~~~~~~~~~^^^^^^^
IndexError: list index out of range
```

Judge reasoning

Test crashed with IndexError during BDD width calculation, preventing reliable results | next: Fix the index out-of-range error in bdd_width function and re-run experiments

---

Matroid Rank Inverse Proportional to ACC^0 Circuit Size for GF(2) Polynomials

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Matroid Theory × ACC^0 Circuit Size
- **Recorded**: 2026-05-10 04:25 UTC
- **Entry ID**: `74050dd8b5ed`

Statement

For any GF(2)-polynomial f over n variables, the rank r of its matroid representation satisfies size_{ACC^0}(f) ≥ Ω(n^{1 + 1/r})

Rationale

Matroid rank captures combinatorial dependencies in GF(2) polynomials, while ACC^0 circuit size reflects algebraic complexity. Their inverse proportionality could reveal structural constraints on circuit depth for dependency-rich functions.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.37s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results; insufficient data to evaluate support fraction or counterexamples | next: Run test with extended timeout and higher resource allocation

---

Secant Variety Dimension Lower Bounds Disjointness Communication Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Secant Variety Geometry × Communication Complexity of DISJOINTNESS
- **Recorded**: 2026-05-10 04:55 UTC
- **Entry ID**: `059c72c72c3c`

Statement

For a random n×n DISJOINTNESS matrix M, the dimension of its secant variety satisfies dim(σ(M)) ≥ Ω(n). This holds for all instances where the communication matrix has no zero rows or columns.

Rationale

Secant variety dimensions capture the algebraic structure of communication matrices. For DISJOINTNESS, the high-dimensional geometry of its matrix (which is a 0-1 matrix with row/column support constraints) may enforce a lower bound on the secant variety's dimension, mirroring the Ω(n) communication complexity lower bound.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_928e789b.py", line 75, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_928e789b.py", line 54, in run_trial
    dim_secant_variety = matrix_rank(M)
                         ^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_928e789b.py", line 41, in matrix_rank
    return gaussian_elimination(A)
           ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_928e789b.py", line 31, in gaussian_elim
    A[j][i] /= factor
```

```
ZeroDivisionError: division by zero
```

Judge reasoning

Test crashed with ZeroDivisionError, preventing evaluation of the conjecture's validity. | next: Fix numerical stability in Gaussian elimination to handle singular matrices

---

Real Critical Point Count Exponential in SOS Degree for Max-CUT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Real Algebraic Geometry × SOS Degree for Max-CUT
- **Recorded**: 2026-05-10 05:06 UTC
- **Entry ID**: `63ccbc681743`

Statement

For a random Max-CUT instance with n variables, the number of real critical points of the quartic polynomial derived from the instance is exponential in the SOS degree d required to achieve an approximation ratio of 0.878 - ε.

Rationale

The number of real critical points reflects the complexity of the optimization landscape. The SOS hierarchy must capture these points to approximate the true minimum, implying a higher degree for more complex landscapes.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1a142126.py", line 123, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1a142126.py", line 98, in run_trial
    critical_points_count = len(critical_points(quartic_poly))
                                ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1a142126.py", line 84, in critical_poin
    n = len(quartic_polynomial)
        ^^^^^^^^^^^^^^^^^^^^^^^
TypeError: object of type 'Fraction' has no len()
```

Judge reasoning

Test crashed due to TypeError in critical points calculation; no data produced to evaluate conjecture | next: Fix the Fraction object handling in critical_points() function and rerun experiments

---

Schatten p-Norm Lower Bound for Disjointness Communication Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Noncommutative L^p Geometry × Communication Complexity of DISJOINTNESS
- **Recorded**: 2026-05-10 05:57 UTC
- **Entry ID**: `835e60ff99be`

Statement

For any n×n matrix M representing a DISJOINTNESS instance, the Schatten p-norm ||M||_p ≥ Ω(√n) for p=2. This bound is tight for the standard DISJOINTNESS matrix and holds for all matrices with rank ≤ log n.

Rationale

Noncommutative L^p norms capture operator-theoretic structure of tensor products, which may encode the entanglement required for communication. The Schatten p=2 norm (Frobenius) relates to the Hilbert-Schmidt energy of the matrix, potentially mirroring the information-theoretic cost of coordination.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `0`, elapsed: `0.03s`

```
unterexample': ''}
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 6.324555320336759, 'instances_tested': 1, '
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 6.324555320336759, 'instances_tested': 1, '
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 6.324555320336759, 'instances_tested': 1, '
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 6.324555320336759, 'instances_tested': 1, '
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 6.324555320336759, 'instances_tested': 1, '
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 6.324555320336759, 'instances_tested': 1, '
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 6.324555320336759, 'instances_tested': 1, '
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 6.324555320336759, 'instances_tested': 1, '
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 6.324555320336759, 'instances_tested': 1, '
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 6.324555320336759, 'instances_tested': 1, '
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 6.324555320336759, 'instances_tested': 1, '
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 6.324555320336759, 'instances_tested': 1, '
TRIAL: {'metric_name': 'Frobenius Norm', 'metric_value': 6.324555320336759, 'instances_tested': 1, '
RESULT: SUPPORTED mean=6.324555320336759 std=0.0 support_fraction=1.0
```

Judge reasoning

Single instance tested (n=40) meets the claimed lower bound exactly, but insufficient to establish asymptotic scaling. The metric value 6.3245 ≈ √40 suggests a constructed test case rather than empirical validation of Ω(√n) for growing n. | next: Test with larger n (e.g., n=1000) to observe scaling behavior and verify the lower bound holds asymptotically

---

SOS Degree Lower Bound via Adjacency Eigenvalue Multiplicity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Real Algebraic Geometry × SOS Refutation Degree for Max-CUT

- **Recorded**: 2026-05-10 06:28 UTC
- **Entry ID**: `8131a890bdd5`

Statement

For any n-vertex graph G, the SOS refutation degree d(G) required to certify Max-CUT(G) ≤ α satisfies d(G) ≥ k(G), where k(G) is the number of distinct positive eigenvalues of G's adjacency matrix. Equality holds for complete bipartite graphs.

Rationale

The Positivstellensatz provides certificates for non-negativity of polynomials, which underlie SOS proofs. The multiplicity of positive eigenvalues reflects the algebraic structure of the quadratic form defining Max-CUT, creating a bridge between spectral geometry and proof complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1dce4a38.py", line 108, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1dce4a38.py", line 89, in run_trial
    eigenvals = eigenvalues(A)
                ^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1dce4a38.py", line 55, in eigenvalues
    Q, R = qr_decomposition(A)
           ^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_1dce4a38.py", line 70, in qr_decomposit
    Q[k] = [v[i] - norm_v * e_k[i] for i in range(n)]
            ~^^^
IndexError: list index out of range
```

Judge reasoning

Test crashed during eigenvalue computation, preventing validation of conjecture | next: Fix the QR decomposition implementation to handle edge cases in eigenvalue calculation

---

Additive Energy Lower Bound via Read-Twice Branching Program Discrepancy

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Additive Combinatorics × Read-Twice Branching Programs
- **Recorded**: 2026-05-10 07:07 UTC
- **Entry ID**: `1de2d7187fe1`

Statement

For every read-twice branching program P computing a function f: $\{0,1\}^n \to \{-1,1\}$, the additive energy E(f) of f is at least $\Omega(n)$ times the discrepancy disc(P) of P.

Rationale

Additive energy measures the number of additive configurations in a function, which could expose structural constraints on functions computable by read-twice BPs. If read-twice BPs have limited additive structure, their discrepancy (a measure of imbalance) might inversely correlate with additive energy, suggesting a nontrivial relationship between algebraic structure and computational complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.07s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Run the test with extended time limits and verify hardware/software constraints

---

Real Root Count Exponential in AC^0 PARITY Circuits

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Real Algebraic Geometry × AC^0 Circuit Size for PARITY
- **Recorded**: 2026-05-10 07:15 UTC
- **Entry ID**: `a1e4626e2b6a`

Statement

For any AC^0 circuit C computing PARITY on n variables, the number of real roots of the polynomial P_C derived from C is $\Theta(2^{n/2})$

Rationale

The parity function's Fourier transform has specific properties, and the number of real roots of P_C could reflect the circuit's structure, potentially exposing lower bounds via real algebraic geometry

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5fd1301b.py", line 88, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5fd1301b.py", line 62, in run_trial
    poly = cnf_to_poly(cnf)
           ^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5fd1301b.py", line 48, in cnf_to_poly
    term *= (1 - x**literal)
                 ^
NameError: name 'x' is not defined
```

Judge reasoning

Test crashed due to undefined variable 'x' in polynomial construction, preventing evaluation of root counts. | next: Fix the polynomial construction code to define variables properly and re-run tests

---

Polymatroid Rank Lower Bound for Monotone CLIQUE

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Polymatroid Theory × Monotone Circuit Size for k-CLIQUE
- **Recorded**: 2026-05-10 07:36 UTC
- **Entry ID**: `834e913154a8`

Statement

For any graph G on n vertices, let M(G) be the polymatroid defined by the rank function r(S) = max{|S ∩ C| : C is a clique in G}. Then r(M(G)) ≥ Ω(n^{1/2}) for all graphs G with clique number ≤ n^{1/2} + 1.

Rationale

Polymatroid rank captures submodular constraints inherent in clique enumeration. By linking this geometric invariant to monotone circuit complexity, we expose structural limitations in representing clique problems with AND/OR gates, potentially revealing new barriers for monotone lower bounds.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.10s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out without producing results, preventing evaluation of support fraction or counterexamples. | next: Run test with extended timeout and higher n values to gather conclusive data

---

Modular Character Degree Gap in Permanent vs Determinant Circuits

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Modular Representation Theory × Monotone Circuit Size for Permanent
- **Recorded**: 2026-05-10 07:48 UTC
- **Entry ID**: `dd4699216f41`

Statement

For a random 3-CNF formula Φ with n variables, let d be its minimal degree of an irreducible representation of the symmetric group S_n over GF(2). Then, the monotone circuit size of the permanent polynomial over GF(2) is at least Ω(2^{d} / n^2).

Rationale

Modular representation theory of symmetric groups provides invariants that could distinguish algebraic structures of permanent and determinant. The degree of irreducible representations relates to symmetry-breaking patterns, which might encode computational complexity via combinatorial constraints on monotone circuits.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.47s`

```
re_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'circuit_size_over_bound', 'metric_value': 0.0, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'circuit_size_over_bound', 'metric_value': 28.125, 'instances_tested': 1, 'co
TRIAL: {'metric_name': 'circuit_size_over_bound', 'metric_value': 0.0, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'circuit_size_over_bound', 'metric_value': 0.0, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'circuit_size_over_bound', 'metric_value': 9.375, 'instances_tested': 1, 'con
TRIAL: {'metric_name': 'circuit_size_over_bound', 'metric_value': 9.375, 'instances_tested': 1, 'con
TRIAL: {'metric_name': 'circuit_size_over_bound', 'metric_value': 12.5, 'instances_tested': 1, 'conj
TRIAL: {'metric_name': 'circuit_size_over_bound', 'metric_value': 0.0, 'instances_tested': 1, 'conje
TRIAL: {'metric_name': 'circuit_size_over_bound', 'metric_value': 0.0, 'instances_tested': 1, 'conje

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_51da1542.py", line 134, in <module>
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_51da1542.py", line 134, in <genexpr>
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
                                   ~^^^^^^^^
KeyError: 'seed'
```

Judge reasoning

Test crashed before completing all trials; incomplete results prevent evaluation of support fraction or counterexamples. | next: Re-run test with debug logging to capture full results and ensure 'seed' metadata is properly recorded.

Class Number Inverse Proportional to Resolution Length in 3-SAT

- **Verdict**: INCONCLUSIVE
- **Bridge**: Algebraic Number Theory × Resolution Proof Length
- **Recorded**: 2026-05-10 08:11 UTC
- **Entry ID**: 3758fc6a2d53

Statement

For a 3-SAT instance with n variables, the class number h(K) of the number field K = Q(√(m)) (where m is the number of clauses) satisfies h(K) ≥ c * (log n) / L, where L is the minimal resolution proof length. Equality holds when the instance is a random 3-SAT formula with density 2.5.

Rationale

Algebraic number theory's class number captures arithmetic complexity of number fields, which may mirror structural properties of SAT instances. The inverse proportionality suggests that harder-to-prove formulas correspond to fields with smaller class numbers, revealing a hidden arithmetic-logical connection.

Novelty

- Judge: INCONCLUSIVE over 10 arXiv hits

Top hits consulted: - [1908.05361v2] Placing quantified variants of 3-SAT and Not-All-Equal 3-SAT in the polynomial hierarchy - [2202.08382v3] The relative class number one problem for function fields, I - [1307.5776v2] An exact algorithm for 1-in-3 SAT - [2402.07935v3] On the Frobenius fields of abelian varieties over number fields - [1803.08437v5] The étale cohomology ring of the ring of integers of a number field

Empirical Test

- exit code: 1, elapsed: 3.56s


```
: 0.2710850306818168, 'instances_tested': 30, 'conjecture_holds': False, 'counterexample': 'Countere
TRIAL: {'metric_name': 'h(K) * L / log(n)', 'metric_value': 0.2710850306818168, 'instances_tested': 3
TRIAL: {'metric_name': 'h(K) * L / log(n)', 'metric_value': 0.2710850306818168, 'instances_tested': 3
TRIAL: {'metric_name': 'h(K) * L / log(n)', 'metric_value': 0.2710850306818168, 'instances_tested': 3
TRIAL: {'metric_name': 'h(K) * L / log(n)', 'metric_value': 0.2710850306818168, 'instances_tested': 3
TRIAL: {'metric_name': 'h(K) * L / log(n)', 'metric_value': 0.2710850306818168, 'instances_tested': 3
TRIAL: {'metric_name': 'h(K) * L / log(n)', 'metric_value': 0.2710850306818168, 'instances_tested': 3
TRIAL: {'metric_name': 'h(K) * L / log(n)', 'metric_value': 0.2710850306818168, 'instances_tested': 3
TRIAL: {'metric_name': 'h(K) * L / log(n)', 'metric_value': 0.2710850306818168, 'instances_tested': 3
TRIAL: {'metric_name': 'h(K) * L / log(n)', 'metric_value': 0.2710850306818168, 'instan
```


Judge reasoning

Test crashed before completing data collection, preventing evaluation of support fraction. Counterexample metric suggests falsification, but incomplete results invalidate definitive conclusion. | next: Re-run test with seed tracking enabled and extended instance count to validate conjecture across complete dataset

---

BIBD Incidence Matrix ACC^0 Circuit Size Lower Bound

- **Verdict**: INCONCLUSIVE
- **Bridge**: Combinatorial Designs (Balanced Incomplete Block Designs) × ACC^0 Circuit Size

- **Recorded**: 2026-05-10 08:23 UTC
- **Entry ID**: `f7b05c9ca4f6`

Statement

For any Balanced Incomplete Block Design (BIBD) with parameters (v, k, λ), the corresponding incidence matrix function f: {0,1}^v → {0,1} requires ACC^0 circuits of size Ω(v^{2 - 1/k}).

Rationale

The regular structure of BIBDs imposes constraints on the incidence matrix that make it resistant to efficient ACC^0 computation, leading to a lower bound on circuit size.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.06s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Increase timeout duration and re-run test with extended computation limits

---

Minimal Incidence Count Inverse Proportional to Disjointness Communication Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Incidence Geometry × Communication Complexity of Disjointness
- **Recorded**: 2026-05-10 09:07 UTC
- **Entry ID**: `bee7b259537e`

Statement

For a disjointness function f on n variables, the minimal number of incidences between the points and lines in the corresponding incidence structure is inversely proportional to the communication complexity of f, i.e., log(communication complexity) = Θ(1 / min_incidence).

Rationale

Incidence geometry provides tools to analyze geometric configurations, which might reveal structural properties of communication matrices that influence their complexity. The inverse proportionality suggests that more structured configurations (higher min_incidence) could imply lower communication complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f4380202.py", line 88, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f4380202.py", line 52, in run_trial
    comm_complexity = compute_communication_complexity(points, lines)
                      ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f4380202.py", line 39, in compute_commu
    if any(point & (1 << k) != 0 for k in points[j]):
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
TypeError: 'int' object is not iterable
```

Judge reasoning

Test crashed due to type error; unable to evaluate conjecture validity | next: Fix data structure handling in compute_communication_complexity to handle integer inputs

---

Standard Young Tableau Count Exponential Gap in Symmetric Power Decompositions

- **Verdict**: INCONCLUSIVE
- **Bridge**: Algebraic Combinatorics (Standard Young Tableaux) × Monotone Circuit Size for Permanent
- **Recorded**: 2026-05-10 09:21 UTC
- **Entry ID**: 04216a9a3c0d

Statement

For every n ≥ 2, the number of standard Young tableaux of shape λ = (n−1,1) for the permanent polynomial exceeds the number for the determinant polynomial by a factor of Ω(2^n).

Rationale

The shape λ = (n−1,1) corresponds to the dominant weight of the symmetric power representation, which captures the combinatorial structure of monomial expansions. The exponential gap in tableau counts reflects the inherent complexity of symmetric vs alternating decompositions in monotone circuits.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_11ee74bb.py", line 48, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
```

```
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_11ee74bb.py", line 31, in run_trial
    ratio = math.log2(T_permanent / T_determinant)
                      ~~~~~~~~~~~~^~~~~~~~~~~~~~~
ZeroDivisionError: division by zero
```

Judge reasoning

Test crashed with ZeroDivisionError, preventing result evaluation. No counterexample found but support fraction cannot be assessed. | next: Fix division-by-zero handling in test code and re-run with diverse seeds

---

Symmetric Group Orbit Count Invariant for AC^0 PARITY Circuits

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Representation Theory of Symmetric Groups × AC^0 circuits computing PARITY
- **Recorded**: 2026-05-10 09:36 UTC
- **Entry ID**: `28c3cf3ded57`

Statement

For any AC^0 circuit C computing PARITY on n variables, the number of distinct irreducible representations of S_n appearing in the decomposition of C's action is Ω(n^{1/2})

Rationale

Symmetric group representations capture combinatorial symmetries of PARITY. The orbit count reflects structural constraints on AC^0 circuits, which cannot exploit high-dimensional symmetry due to depth limitations.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.39s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing evaluation of support/falsification criteria | next: Increase time limit and re-run test with optimized code for larger n values

---

Free Entropy Distinguishes Read-Twice BPs from IP_2 Trivial Ones

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Free Probability × Read-Twice Branching Programs
- **Recorded**: 2026-05-10 09:46 UTC
- **Entry ID**: `eea147175457`

Statement

For a read-twice branching program P over n variables, let M_P be the matrix whose entries are the free entropy of the adjacency matrix of P's layers. Then, the minimum eigenvalue of M_P is Θ(log size(P)), while for the IP_2 trivial BP, it is Ω(n).

Rationale

Free entropy captures spectral properties of random matrices, which may reveal structural differences between read-twice BPs (with logarithmic size) and the IP_2 trivial BP (with linear size). The tensor product structure of M_P ensures computable invariants.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.13s`

```
alue': 1, 'instances_tested': 1, 'conjecture_holds': False, 'counterexample': 'IP_2 trivial BP'}
TRIAL: {'metric_name': 'min_eigenvalue', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds
TRIAL: {'metric_name': 'min_eigenvalue', 'metric_value': 2, 'instances_tested': 1, 'conjecture_holds
TRIAL: {'metric_name': 'min_eigenvalue', 'metric_value': 4, 'instances_tested': 1, 'conjecture_holds
TRIAL: {'metric_name': 'min_eigenvalue', 'metric_value': 0, 'instances_tested': 1, 'conjecture_holds
TRIAL: {'metric_name': 'min_eigenvalue', 'metric_value': 5, 'instances_tested': 1, 'conjecture_holds
TRIAL: {'metric_name': 'min_eigenvalue', 'metric_value': 2, 'instances_tested': 1, 'conjecture_holds
TRIAL: {'metric_name': 'min_eigenvalue', 'metric_value': 2, 'instances_tested': 1, 'conjecture_holds
TRIAL: {'metric_name': 'min_eigenvalue', 'metric_value': 1, 'instances_tested': 1, 'conjecture_holds
TRIAL: {'metric_name': 'min_eigenvalue', 'metric_value': 0, 'instances_tested': 1, 'conjecture_holds

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_bc1e3e6a.py", line 105, in <module>
    first_failing_seed = next(res["seed"] for res in results if not res["conjecture_holds"])
                              ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_bc1e3e6a.py", line 105, in <genexpr>
    first_failing_seed = next(res["seed"] for res in results if not res["conjecture_holds"])
                                       ~~~^^^^^^^^
KeyError: 'seed'
```

Judge reasoning

Test crashed before producing reliable data, preventing validation of the conjecture despite apparent counterexamples. | next: Fix the test's KeyError by implementing proper seed tracking and re-run with sufficient samples

---

Association Scheme Eigenvalue Inverse Proportional to ACC^0 Circuit Size for CNF Formulas

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Association Schemes × ACC^0 Circuit Size for CNF Formulas
- **Recorded**: 2026-05-10 09:56 UTC
- **Entry ID**: `680124d1df5f`

Statement

For any CNF formula with n variables, the size of the minimal ACC^0 circuit computing it is inversely proportional to the absolute value of the smallest eigenvalue of the association scheme constructed from the formula's clause structure.

Rationale

Association schemes capture combinatorial regularities that might reflect the structure of CNF formulas, and their eigenvalues could encode information about the complexity of computing the function. By linking eigenvalues to circuit size, we might uncover new structural insights.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.04s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Run test with extended timeout and targeted CNF instances exhibiting potential eigenvalue-circuit size correlations

---

Schatten p-Norm Lower Bound for Disjointness Communication Matrices

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Noncommutative L^p Geometry × Communication Complexity of DISJOINTNESS
- **Recorded**: 2026-05-10 10:35 UTC
- **Entry ID**: `9070a0d44287`

Statement

For the communication matrix M of the DISJOINTNESS problem on n elements, the Schatten p-norm ||M||_p satisfies ||M||_p ≥ Ω(n^{1/2 - 1/p}) for all p ∈ [2, ∞). This bound is tight up to polylog(n) factors for p = 2 and p = log n.

Rationale

Schatten norms capture operator-theoretic structure of matrices, while DISJOINTNESS requires Ω(n) communication. By analyzing how matrix norms scale with n, we may uncover geometric constraints on information-theoretic lower bounds. The conjecture leverages the interplay between noncommutative geometry and communication complexity via matrix/operator norms.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 1.35s


```
0, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'Schatten p-norm ratio', 'metric_value': 7002758039529.397, 'instances_tested
TRIAL: {'metric_name': 'Schatten p-norm ratio', 'metric_value': 5366495637037.515, 'instances_tested
TRIAL: {'metric_name': 'Schatten p-norm ratio', 'metric_value': 4677308384809.626, 'instances_tested
TRIAL: {'metric_name': 'Schatten p-norm ratio', 'metric_value': 6724048925014.629, 'instances_tested
TRIAL: {'metric_name': 'Schatten p-norm ratio', 'metric_value': 5222112081379.986, 'instances_tested
TRIAL: {'metric_name': 'Schatten p-norm ratio', 'metric_value': 4310746725920.3105, 'instances_teste
TRIAL: {'metric_name': 'Schatten p-norm ratio', 'metric_value': 3762614997373.867, 'instances_tested
TRIAL: {'metric_name': 'Schatten p-norm ratio', 'metric_value': 1649336149815.1272, 'instances_teste
TRIAL: {'metric_name': 'Schatten p-norm ratio', 'metric_value': 4567815424813.691, 'instances_tested
TRIAL: {'metric_name': 'Schatten p-norm ratio', 'metric_value': 5347078012104.417, 'instances_tested
TRIAL: {'metric_name': 'Schatten p-norm ratio', 'metric_value': 6400800002636.159, 'instances_tested
TRIAL: {'metric_name': 'Schatten p-norm ratio', 'metric_value': 7769185482545.169, 'instances_tested
RESULT: SUPPORTED mean=5289924136849.4082 std=1369787444112.6799 support_fraction=1.00
```


Judge reasoning

Safety rail: critic_challenge | original judge: All 12 trials confirm the conjecture holds with 100% support, exceeding the 80% threshold. The metric values consistently align with the Ω(n^{1/2 - 1/ | next: Test the conjecture for p approaching ∞ and verify the polylog(n) tightness claim for p=2

Free Entropy Gap in Read-Twice Branching Programs for IP_2

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Free Probability × Read-Twice Branching Programs
- **Recorded**: 2026-05-10 11:14 UTC
- **Entry ID**: `0b9d1a98a4f1`

Statement

For a read-twice branching program P computing IP_2 (inner product mod 2), let μ_P be the empirical spectral distribution of its transition matrix. The free entropy ϕ(μ_P) = -∫ log|z| dμ_P(z) satisfies ϕ(μ_P) = Θ(log n) for trivial P (e.g., constant functions), but ϕ(μ_P) ≥ Ω(n) for nontrivial P. The gap is witnessed by the spectral radius λ_P = max{|z| : z ∈ spec(μ_P)} with λ_P ≥ 1 + Ω(1/n) for nontrivial P.

Rationale

Free entropy quantifies the 'spread' of spectral measures, capturing structural complexity in noncommutative probability spaces. Read-twice BPs for IP_2 exhibit distinct spectral signatures: trivial programs concentrate measure near 1, while nontrivial ones spread it across eigenvalues, creating a detectable free entropy gap.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dc3e2548.py", line 89, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dc3e2548.py", line 66, in run_trial
    eigenvalues = spectral_radius(transition_matrix)
                  ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dc3e2548.py", line 45, in spectral_radi
    v /= sum(v)
TypeError: unsupported operand type(s) for /=: 'list' and 'float'
```

Judge reasoning

Test crashed during spectral radius computation, preventing data collection. No results available to assess conjecture. | next: Implement numerical stability fixes for spectral radius calculation and re-run experiments

---

Non-Commutative Fourier L1 Norm Distinguishes Read-Twice from Read-Once BPs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Non-Commutative Harmonic Analysis × Read-Twice Branching Programs
- **Recorded**: 2026-05-10 11:23 UTC
- **Entry ID**: `31bf8e50927c`

Statement

For any read-twice branching program P over n variables, the L1 norm of its non-commutative Fourier transform satisfies $||F(P)||_1 \geq n/2$, whereas for read-once BPs, $||F(P)||_1 = O(\log n)$.

Rationale

Non-commutative Fourier analysis captures branching program structure via group representations, exposing inherent symmetry gaps between read-twice and read-once models. The L1 norm quantifies this asymmetry through harmonic decomposition.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_df9169c3.py", line 80, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_df9169c3.py", line 51, in run_trial
    F_ro = discrete_fourier_transform(read_once_bp)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_df9169c3.py", line 28, in discrete_four
    omega = [math.exp(-2j * math.pi * i / n) for i in range(n)]
```

```
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
TypeError: must be real number, not complex
```

Judge reasoning

Test crashed due to type error in Fourier transform implementation, preventing validation of conjecture | next: Fix the discrete_fourier_transform function to handle complex numbers properly and re-run tests

---

Automorphism Group Generator Count Bounded by ABP Width for Symmetric CNFs

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Group Theory (Permutation Group Generators) × Algebraic Branching Programs (ABP Width)
- **Recorded**: 2026-05-10 11:43 UTC
- **Entry ID**: `e9123a8ea921`

Statement

For any symmetric CNF formula Φ with automorphism group G, the minimal width of an ABP representing Φ is exactly the minimal number of generators required to generate G.

Rationale

Symmetric CNFs have automorphism groups that capture their symmetry. The minimal generating set size of G reflects the structural complexity of the symmetry, which could influence the minimal AB. width needed to represent the formula, as ABPs are tied to permutation-based computations.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.10s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Increase timeout duration and re-run test with extended computation limits

---

Jordan Algebra Rank Inverse Proportional to ABP Width for Symmetric Functions

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Jordan Algebra Theory × Algebraic Branching Program Width
- **Recorded**: 2026-05-10 11:59 UTC
- **Entry ID**: `30850b31662a`

Statement

For any symmetric Boolean function f on n variables, the Jordan algebra rank of its symmetric polynomial representation is inversely proportional to the minimal ABP width required to compute f. Specifically, rank(f) = Θ(n / width(f))

Rationale

The Jordan algebra structure imposes constraints on the symmetric polynomial's representation, limiting the required width of ABPs. This could reveal algebraic constraints on the function's complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.39s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Retry test with extended timeout and debug logs to identify crash cause

---

Matroid Rank Inverse Proportional to Monotone DNF Discrepancy for k-CLIQUE

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Matroid Theory × Monotone DNF Discrepancy
- **Recorded**: 2026-05-10 12:05 UTC
- **Entry ID**: `70db22789577`

Statement

For a k-CLIQUE instance, the discrepancy of its monotone DNF representation is Θ(1/rank(M)), where M is the vector matroid over GF(2) derived from the incidence matrix of the formula's clauses.

Rationale

Matroid rank captures the combinatorial structure of clause dependencies, while discrepancy quantifies the uniformity of clause coverage. Their inverse proportionality suggests inherent complexity barriers for monotone circuits.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5045b20c.py", line 86, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5045b20c.py", line 60, in run_trial
    rank = gaussian_elimination(incidence_matrix)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5045b20c.py", line 31, in gaussian_elim
    factor = Fraction(M[j][i], M[i][i])
             ^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/fractions.py", line 281, in __new__
    raise ZeroDivisionError('Fraction(%s, 0)' % numerator)
ZeroDivisionError: Fraction(0, 0)
```

Judge reasoning

Test crashed with ZeroDivisionError during Gaussian elimination, preventing result validation. | next: Inspect incidence matrix construction to resolve division-by-zero error in Gaussian elimination

---

Symmetric Group Fourier Coefficient Sum Exponential Gap Between Permanent and Determinant

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Representation Theory of Symmetric Groups × Boolean Circuit Complexity (Permanent vs Determinant)
- **Recorded**: 2026-05-10 12:27 UTC
- **Entry ID**: `e58723759ed4`

Statement

For any n ≥ 2, the sum of absolute values of Fourier coefficients of the permanent function over S_n exceeds that of the determinant function by Ω(2^{n/2}), where coefficients are computed using the Young symmetrizer decomposition of the group algebra.

Rationale

The permanent's positivity under symmetric group actions creates constructive interference in Fourier coefficients, while determinant's alternating signs induce destructive interference. This gap mirrors the orbit closure separation in GCT, offering a combinatorial proxy for algebraic complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2665b4a7.py", line 148, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
```

```
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2665b4a7.py", line 120, in run_trial
    perm_basis = young_symmetrizer((n,), 1)
                 ^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2665b4a7.py", line 56, in young_symmetr
    for cell in range(len(row)):
                      ^^^^^^^^
TypeError: object of type 'int' has no len()
```

Judge reasoning

Test crashed due to TypeError in young_symmetrizer function, preventing data collection | next: Fix the young_symmetrizer function to handle integer inputs properly and re-run experiments

---

Schur-Weyl Duality Invariant for Permanent vs Determinant Circuit Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Schur-Weyl Duality × Boolean Circuit Complexity (Permanent vs Determinant)
- **Recorded**: 2026-05-10 12:38 UTC
- **Entry ID**: `0032e472e3ec`

Statement

For a homogeneous polynomial f of degree n in n variables, the number of standard Young tableaux in its decomposition under GL(n) is exponential in n for the permanent and polynomial in n for the determinant.

Rationale

Schur-Weyl duality reveals structural differences in the representation theory of the permanent and determinant, which may imply inherent complexity gaps. The number of tableaux could reflect the number of terms in the decomposition, directly relating to circuit size.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `-9`, elapsed: `103.92s`

```
(no output)
```

Judge reasoning

Test crashed before producing data, preventing evaluation of support fraction or counterexamples. | next: Re-run test with increased resource limits and debug logging to identify crash cause

---

Schur-Weyl Dimension Gap in 3-CNF Permanent vs Determinant Shapes

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Schur-Weyl Duality × 3-CNF SAT Instances
- **Recorded**: 2026-05-10 12:46 UTC
- **Entry ID**: `390e4c06544a`

Statement

For random 3-CNF instances with n variables and m clauses, the number of standard Young tableaux of shape (m, m, ..., m) (m rows of length 3) is ≥ $2^{\{n}2/4\}$ times the number for shape (n, n-1, ..., 1).

Rationale

Schur-Weyl duality links polynomial representations to symmetric group actions; the dimension gap between rectangular and staircase Young diagrams mirrors the algebraic complexity of permanent vs determinant. Exponential separation in SYT counts could imply inherent structural differences in their representation-theoretic decompositions.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c300c5c1.py", line 115, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c300c5c1.py", line 91, in run_trial
    ratio = permanent_syt_count / (2 ** (n ** 2 / 4) * determinant_syt_count)
            ~~~~~~~~~~~~~~~~~~~~^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
ZeroDivisionError: float division by zero
```

Judge reasoning

Test crashed with ZeroDivisionError, preventing validation of conjecture. Likely indicates code error rather than mathematical failure. | next: Debug the division-by-zero error in the test code's ratio calculation

---

Noncommutative L^p Norm Lower Bound for Disjointness Communication Matrices

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Noncommutative L^p Geometry × Communication Complexity of DISJOINTNESS
- **Recorded**: 2026-05-10 13:07 UTC
- **Entry ID**: `b70c19d6f5c7`

Statement

For all n ≥ 1, the Schatten p-norm of the DISJOINTNESS communication matrix M_n satisfies ||M_n||_p ≥ Ω(n^{1/p}) for p ∈ {2, 4, 8}, with equality achieved by the uniform distribution over disjointness instances.

Rationale

Noncommutative L^p norms capture operator-level structure of matrices, which may reveal hidden geometric constraints in communication matrices. The conjecture links p-norm scaling to communication complexity via operator space duality, bypassing traditional rank-based methods.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 0.14s

```
'n=40, p=2: 6.164414002968976, p=4: 2.4828237961983883, p=8: 1.575697875926216'}
TRIAL: {'metric_name': 'Schatten_p_norm', 'metric_value': 2.3992288909229176, 'instances_tested': 90
TRIAL: {'metric_name': 'Schatten_p_norm', 'metric_value': 2.435291859336928, 'instances_tested': 90,
TRIAL: {'metric_name': 'Schatten_p_norm', 'metric_value': 2.450571445248159, 'instances_tested': 90,
TRIAL: {'metric_name': 'Schatten_p_norm', 'metric_value': 2.4639221878698567, 'instances_tested': 90
TRIAL: {'metric_name': 'Schatten_p_norm', 'metric_value': 2.476655292912502, 'instances_tested': 90,
TRIAL: {'metric_name': 'Schatten_p_norm', 'metric_value': 2.4245870039599913, 'instances_tested': 90
TRIAL: {'metric_name': 'Schatten_p_norm', 'metric_value': 2.4603378015952155, 'instances_tested': 90
TRIAL: {'metric_name': 'Schatten_p_norm', 'metric_value': 2.3885132175919916, 'instances_tested': 90
TRIAL: {'metric_name': 'Schatten_p_norm', 'metric_value': 2.4469539486632144, 'instances_tested': 90
```

Judge reasoning

parse_fail | next: NONE

Schur Coefficient Gap in Permanent vs Determinant Decompositions

- **Verdict**: BARRIER_HIT
- **Bridge**: Representation Theory of Symmetric Groups × Boolean Circuit Complexity
- **Recorded**: 2026-05-10 13:12 UTC
- **Entry ID**: 94821ab926f3

Statement

For all n and m < n^{1.5}, the minimal Schur coefficient in the decomposition of the permanent polynomial perm_n is exponentially larger than that of the determinant polynomial det_m. This gap implies that the minimal circuit size for perm_n exceeds that of det_m by a factor of $\Omega(2^{\{n}\varepsilon\})$ for any $\varepsilon > 0$.

Rationale

Schur coefficients encode the symmetric structure of polynomials, and their exponential disparity between perm_n and det_m could reflect inherent complexity differences. This might expose algebraic barriers to efficient computation, aligning with GCT's orbit closure separation goals.

Novelty

- Judge: “ over 0 arXiv hits

Empirical Test

- exit code: -1, elapsed: 0.00s

```
(no output)
```

Judge reasoning

Rejected by ALGEBRIZATION barrier

Secant Variety Dimension Gap in Determinant vs Permanent Orbits for SAT Instances

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic Geometry (Secant Varieties) × SAT Instances
- **Recorded**: 2026-05-10 14:15 UTC
- **Entry ID**: `61a6865ba577`

Statement

For a random 3-CNF formula Φ with n variables, the dimension of the secant variety of the determinant's orbit closure is at least 2^n/2 smaller than that of the permanent's orbit closure. This gap is quantified by the difference in the number of equations defining their ideals over C.

Rationale

Secant varieties capture the complexity of polynomial families via their geometric structure. The determinant's orbit has a well-understood secant variety, while the permanent's lacks such analysis. A dimension gap could reflect inherent complexity differences, with SAT instances providing concrete test cases for this algebraic-geometric hypothesis.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.10s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Increase timeout duration and re-run with stricter resource limits to capture full execution

---

Fourier Coefficient Discrepancy and ACC^0 Circuit Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Additive Combinatorics × ACC^0 Circuit Size
- **Recorded**: 2026-05-10 14:31 UTC
- **Entry ID**: `de5fe0b83dbc`

Statement

For a Boolean function f derived from a 3-CNF formula on n variables, the discrepancy of its Fourier coefficients (defined as max_k |F(k)| - min_k |F(k)|) is at least Ω(2^{n/2}) if f requires ACC^0 circuits of size ≥ 2^{n/2}.

Rationale

Additive combinatorics provides tools to analyze structured distributions of Fourier coefficients, which may expose inherent complexity barriers for ACC^0 circuits. Discrepancy captures non-uniformity in Fourier spectra, potentially correlating with circuit depth.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9b6cc439.py", line 85, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9b6cc439.py", line 64, in run_trial
    f = fourier_coefficients(clauses, n)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9b6cc439.py", line 42, in fourier_coeff
    f = [0] * (1 << num_vars)
        ~~~~^~~~~~~~~~~~~~~~~
MemoryError
```

Judge reasoning

Test crashed with MemoryError, preventing evaluation of the conjecture's validity. | next: Optimize memory usage for large n or increase hardware limits to retest

---

Fourier Coefficient Concentration and ACC^0 Circuit Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Fourier Analysis of Boolean Functions × ACC^0 Circuit Size for 3-CNF Formulas
- **Recorded**: 2026-05-10 14:39 UTC
- **Entry ID**: `56779c9244c7`

Statement

For a 3-CNF formula with n variables, the sum of absolute Fourier coefficients is at least Ω(2^{n/2}) if the formula cannot be computed by an ACC^0 circuit of size o(2^{n/2}).

Rationale

Fourier coefficients capture variable influence and correlation structure, which are critical for ACC^0 lower bounds. High concentration in Fourier coefficients may indicate structured functions resistant to ACC^0 computation.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2c6c15b2.py", line 74, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2c6c15b2.py", line 49, in run_trial
    f = [0] * (1 << n)
        ~~~~^~~~~~~~~~
MemoryError
```

Judge reasoning

Test crashed with MemoryError, preventing data collection. Cannot assess conjecture validity without successful execution. | next: Optimize memory usage in the Fourier coefficient computation (e.g., use sparse representations or iterative methods)

Moment Matrix Rank Lower Bound for Max-CUT SOS Approximation

- **Verdict**: INCONCLUSIVE
- **Bridge**: Real Algebraic Geometry × SOS Degree for Max-CUT
- **Recorded**: 2026-05-10 14:54 UTC
- **Entry ID**: b1a344420349

Statement

For any n-vertex graph with max-CUT value ≥ $0.878n^2$, the rank of its degree-d SOS moment matrix is ≥ Ω(n) if d < log n / log log n.

Rationale

The moment matrix's rank captures the algebraic complexity of polynomial certificates for max-CUT. A lower bound on this rank implies that the SOS hierarchy requires super-logarithmic depth to approximate the cut, revealing structural constraints on polynomial identities.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e6a56c87.py", line 108, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e6a56c87.py", line 87, in run_trial
    poly = max_cut_polynomial(graph)
           ^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e6a56c87.py", line 33, in max_cut_polyn
    poly += f"x{i} * x{j}"
```

```
TypeError: unsupported operand type(s) for +=: 'int' and 'str'
```

Judge reasoning

Test crashed due to type error before producing results; no data to evaluate conjecture | next: Debug the polynomial construction code to fix the int-str TypeError

---

Kronecker Coefficient Non-Zero Threshold for Permanent-Complete SAT Instances

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Representation Theory of Symmetric Groups × SAT Instances
- **Recorded**: 2026-05-10 15:10 UTC
- **Entry ID**: `c93b44a7d21e`

Statement

For a random 3-CNF formula Φ with n variables and m clauses, let $\lambda = (\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_k)$ be the partition encoding the clause-variable incidence matrix's row sums. Define k(Φ) = ∑_{i=1}^k λ_i. Then, the Kronecker coefficient g(λ, μ, ν) for partitions μ = (n), ν = (m) is non-zero iff Φ is unsatisfiable, and k(Φ) ≤ 2n log n.

Rationale

Kronecker coefficients govern tensor product decompositions of symmetric group representations. By encoding SAT instances as incidence matrices, their combinatorial structure maps to partition tuples. Non-zero Kronecker coefficients imply representation-theoretic entanglement that may correlate with computational hardness, offering a bridge between algebraic combinatorics and SAT complexity.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.10s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2c8df4b8.py", line 122, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2c8df4b8.py", line 98, in run_trial
    clauses = generate_3cnf(n, m)
              ^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2c8df4b8.py", line 21, in generate_3cnf
    clause = random.sample(variables, 3)
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/random.py", line 413, in sample
    raise TypeError("Population must be a sequence.  "
TypeError: Population must be a sequence.  For dicts or sets, use sorted(d).
```

Judge reasoning

Test crashed due to invalid population type in random.sample; unable to evaluate conjecture's validity | next: Fix generate_3cnf() to use ordered sequences (e.g., lists) for variables and re-run tests

---

SOS Degree Lower Bound via Matroid Polytope Dimension

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Combinatorial Geometry (Matroid Polytopes) × SOS Refutation Degree for CSPs
- **Recorded**: 2026-05-10 16:28 UTC
- **Entry ID**: `b4a7c853b857`

Statement

For a CSP instance represented as a hypergraph H, the SOS refutation degree required to prove unsatisfiability is at least the dimension of the matroid polytope formed by the characteristic vectors of H's hyperedges. Formally, dim(polytope(H)) ≤ deg_SOS(H) for all hypergraphs H with n ≤ 40.

Rationale

Matroid polytopes encode combinatorial constraints via convex hulls, while SOS degrees measure algebraic complexity of CSPs. Their direct linkage could reveal geometric obstructions to polynomial-time refutations, leveraging the interplay between discrete geometry and algebraic optimization.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `247.53s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Increase timeout duration and re-run test with extended computation limits

---

Schur Coefficient Exponential Gap in 3-CNF Permanent vs Determinant

- **Verdict**: `BARRIER_HIT`
- **Bridge**: Representation Theory of Symmetric Groups × Boolean Circuit Size for 3-CNF Formulas
- **Recorded**: 2026-05-10 16:34 UTC
- **Entry ID**: `c87370f8ed72`

Statement

For random 3-CNF formulas with n variables, the sum of absolute Schur coefficients for the permanent polynomial exceeds that of the determinant by Ω(2^{n/2}), where Schur coefficients are computed via the decomposition of the clause incidence tensor into symmetric powers.

Rationale

Schur coefficients capture symmetry-breaking patterns in tensor rank. Permanent's higher symmetry complexity (via Young tableaux) creates a gap detectable via character orthogonality, while determinant's structure limits its coefficient growth under linear substitutions.

Novelty

- Judge: “ over 0 arXiv hits

Empirical Test

- exit code: `-1`, elapsed: `0.00s`

```
(no output)
```

Judge reasoning

Rejected by ALGEBRIZATION barrier

Symmetric Group Fourier Coefficient Gap in Disjointness Communication Matrices

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Representation Theory of Symmetric Groups × Communication Complexity of DISJOINTNESS
- **Recorded**: 2026-05-10 17:22 UTC
- **Entry ID**: `7003ce43815a`

Statement

For any n×n disjointness matrix M, the maximum Fourier coefficient |χ_λ(M)| over irreducible symmetric group characters χ_λ satisfies |χ_λ(M)| ≥ n^2/2 for λ = (n-1,1) and |χ_λ(M)| ≤ n/2 for λ = (n).

Rationale

Symmetric group representations capture combinatorial symmetries of disjointness matrices. The (n-1,1) partition corresponds to transpositions, which encode pairwise conflicts central to disjointness. The (n) partition corresponds to the trivial representation, which reflects global consistency. The conjecture links representation-theoretic structure to communication complexity via Fourier analysis.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c28d86c4.py", line 94, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
```

```
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c28d86c4.py", line 77, in run_trial
    chi_lambda_1 = fourier_coefficient(M, lambda_1)
                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c28d86c4.py", line 65, in fourier_coeff
    char_table = character_table(n)
                 ^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c28d86c4.py", line 58, in character_tab
    char_table_n_minus_k_l = character_table(n - k)[j][j]
                             ^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c28d86c4.py", line 58, in character_tab
    char_table_n_minus_k_l = character_table(n - k)[j][j]
                             ^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c28d86c4.py", line 58, in character_tab
    char_table_n_minus_k_l = character_table(n - k)[j][j]
                             ^^^^^^^^^^^^^^^^^^^^^^
  [Previous line repeated 10 more times]
IndexError: list index out of range
```

Judge reasoning

Test crashed with IndexError before completing; unable to verify conjecture's validity | next: Fix the character table generation to handle index ranges properly and re-run tests

---

Harmonic Coefficient Sum Lower Bound for Resolution Proof Length

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Harmonic Analysis on Finite Groups × Resolution Proof Length
- **Recorded**: 2026-05-10 17:37 UTC
- **Entry ID**: `161db2211085`

Statement

For any 3-SAT instance with n variables, the resolution proof length is at least the sum of absolute values of its Fourier coefficients over the hypercube group {0,1}^n. Formally: length(Π) ≥ ∑_{S⊆[n]} |f̂(S)|, where f is the indicator function of satisfying assignments.

Rationale

Harmonic analysis reveals the function's structure via Fourier coefficients, which may correlate with proof complexity. The sum of absolute coefficients quantifies the function's 'spread' over the hypercube, potentially reflecting the number of clauses needed to resolve.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `124`, elapsed: `240.20s`

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Increase timeout duration and re-run tests with diverse seed configurations

---

Noncommutative L^p Norm Lower Bounds for Disjointness Communication Matrices

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Noncommutative L^p Geometry × Communication Complexity of DISJOINTNESS
- **Recorded**: 2026-05-10 18:03 UTC
- **Entry ID**: `d090c650622d`

Statement

For any n×n matrix M representing a DISJOINTNESS instance, the noncommutative L^p norm ||M||_{p} ≥ Ω(√n) for p ∈ (1,2). This holds for all matrices where the support is a product set S × T with |S| = |T| = n^{1/2}.

Rationale

Noncommutative L^p norms capture operator-theoretic structure of tensor products, which may encode geometric obstructions to efficient communication. The product support constraint ensures the matrix lies in the intersection of a secant variety and a noncommutative convex body, creating a natural test case for norm-communication tradeoffs.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d597251a.py", line 56, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d597251a.py", line 34, in run_trial
    u = [M_scaled[i][j] * v[j] for j in range(n)]
                  ^
UnboundLocalError: cannot access local variable 'i' where it is not associated with a value
```

Judge reasoning

Test crashed due to UnboundLocalError, preventing data collection. Pre-registered support condition cannot be evaluated without valid results. | next: Fix the code bug in test_d597251a.py line 34 (unbound 'i' variable) and re-run experiments

---

Cheeger Constant Exponentiates Resolution Length for Tseitin Formulas

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Spectral Geometry × Resolution Proof Length

- **Recorded**: 2026-05-10 19:08 UTC
- **Entry ID**: `933eeb640129`

Statement

For a Tseitin formula derived from a d-regular graph G, the Resolution proof length is at least $2^{\Omega(h(G))}$, where h(G) is the Cheeger constant of G.

Rationale

Expanders have high Cheeger constants, which correlate with the difficulty of Resolution proofs. The spectral properties of the graph influence the structure of the proof, leading to exponential lower bounds.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7536d2ac.py", line 116, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7536d2ac.py", line 100, in run_trial
    length = resolution_length(graph)
             ^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7536d2ac.py", line 83, in resolution_le
    while not dpll():
              ^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7536d2ac.py", line 56, in dpll
    if not clauses:
           ^^^^^^^
UnboundLocalError: cannot access local variable 'clauses' where it is not associated with a value
```

Judge reasoning

Test crashed with UnboundLocalError, preventing data collection. The error suggests a code implementation flaw in the DPLL solver. | next: Fix the DPLL implementation to handle clause initialization properly and re-run the experiment

---

Hilbert Function Inverse Proportional to 3-SAT Solution Count

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Commutative Algebra (Hilbert Function of Ideals) × 3-SAT Instance Solution Count
- **Recorded**: 2026-05-10 19:52 UTC
- **Entry ID**: `b455e558d6c5`

Statement

For a random 3-SAT instance with n variables, the Hilbert function $H_I(d)$ of the ideal I generated by the clauses satisfies $H_I(n/2) = \Theta(1 / (\text{number of solutions}))$, where the constant depends on n.

Rationale

The algebraic structure of the ideal reflects the combinatorial complexity of the SAT instance, with the Hilbert function capturing the dimension of the solution space's dual.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_acd903d9.py", line 108, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_acd903d9.py", line 75, in run_trial
    solutions_count = dpll(sat_formula)
                      ^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_acd903d9.py", line 64, in dpll
    return backtrack({})
           ^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_acd903d9.py", line 53, in backtrack
    result = backtrack(model)
             ^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_acd903d9.py", line 53, in backtrack
    result = backtrack(model)
             ^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_acd903d9.py", line 58, in backtrack
    result = backtrack(model)
             ^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_acd903d9.py", line 58, in backtrack
    result = backtrack(model)
             ^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_acd903d9.py", line 58, in backtrack
    result = backtrack(model)
             ^^^^^^^^^^^^^^^^
   [Previous line repeated 990 more times]
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_acd903d9.py", line 53, in backtrack
    result = backtrack(model)
             ^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_acd903d9.py", line 50, in backtrack
    return solve(model)
           ^^^^^^^^^^^^
RecursionError: maximum recursion depth exceeded
```

Judge reasoning

Test crashed with recursion depth error, preventing data collection. No counterexample found but insufficient evidence to support conjecture. | next: Implement recursion limit increases or switch to iterative backtracking for deeper formula evaluation

---

Tropical Rank Gap in Read-Twice BP Transition Matrices

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Tropical Linear Algebra × Read-Twice Branching Programs
- **Recorded**: 2026-05-10 20:25 UTC
- **Entry ID**: `965a23f70cf0`

Statement

For any read-twice BP over GF(2) with n variables, the tropical rank of its transition matrix is Ω(n), while read-once BPs have tropical rank O(log n).

Rationale

Tropical rank captures the minimal number of rank-1 tropical matrices needed to decompose a matrix, reflecting structural complexity. Read-twice BPs, with their repeated state access, induce higher tropical rank due to entangled state transitions, contrasting with read-once BPs' decomposable structure.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5231202d.py", line 157, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5231202d.py", line 131, in run_trial
    A = generate_bp_transition_matrix(clauses, n, p)
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5231202d.py", line 110, in generate_bp_
      if any(all((s >> (v-1) & 1) == (t >> (abs(v)-1) & 1)) or all((s >> (v-
1) & 1) != (t >> (abs(v)-1) & 1)) for v in range(1, n+1)):
       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5231202d.py", line 110, in <genexpr>
      if any(all((s >> (v-1) & 1) == (t >> (abs(v)-1) & 1)) or all((s >> (v-
1) & 1) != (t >> (abs(v)-1) & 1)) for v in range(1, n+1)):
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
TypeError: 'bool' object is not iterable
```

Judge reasoning

Test crashed due to TypeError, preventing data collection. Pre-registered support condition cannot be evaluated without successful trials. | next: Debug the generate_bp_transition_matrix function's boolean expression handling

---

SOS Moment Matrix Rank and Max-CUT Integrality Gap

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Real Algebraic Geometry × SOS Degree for Max-CUT

- **Recorded**: 2026-05-10 20:31 UTC
- **Entry ID**: `8d60c087478f`

Statement

For a random Max-CUT instance on n variables, the rank of the moment matrix in the d-th level of the SOS hierarchy is at least $\Omega(n/d^2)$, and this bound is tight for instances with high expansion properties.

Rationale

The rank of the moment matrix captures the complexity of the polynomial constraints. High expansion in Max-CUT instances may require higher rank to capture the necessary inequalities, linking algebraic structure to SOS degree.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7f687d6d.py", line 90, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7f687d6d.py", line 73, in run_trial
    M = moment_matrix(G, d)
        ^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7f687d6d.py", line 39, in moment_matrix
    M[n+k-1][n+j] = -1
    ~~~~~~~~^^^^^
IndexError: list assignment index out of range
```

Judge reasoning

Test crashed with IndexError, preventing data collection. The pre-registered support condition cannot be evaluated without valid results. | next: Debug the moment_matrix implementation to fix index out-of-range errors and re-run tests

---

Genus of Communication Curve Bounds Disjointness Complexity

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Algebraic Geometry (Curves) × Communication Complexity of DISJOINTNESS
- **Recorded**: 2026-05-10 20:37 UTC
- **Entry ID**: `e10a38fc2892`

Statement

For any n×n disjointness matrix M over GF(2), the genus g of the curve defined by the polynomial f(x,y) = ∑_{i,j} M[i][j]$x^{\{i\}y}${j} satisfies g ≥ log_2(1 + 2^{n/2}) - 2. This bound is tight for the standard disjointness instance.

Rationale

The genus captures topological complexity of the curve, which reflects the algebraic structure of the communication matrix. Disjointness requires separating algebraic varieties, where genus serves as a natural measure of 'interleaving complexity'.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c1bea3de.py", line 103, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c1bea3de.py", line 82, in run_trial
    g = compute_genus(f_poly)
        ^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c1bea3de.py", line 73, in compute_genus
    det_C = gaussian_elimination(C, [0] * (n + 1))
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c1bea3de.py", line 47, in gaussian_elim
    M[i][j] /= factor
ZeroDivisionError: division by zero
```

Judge reasoning

Test crashed with ZeroDivisionError, preventing evaluation of the conjecture's validity | next: Fix the Gaussian elimination implementation to handle zero pivots or verify matrix construction logic

---

Toric Variety Hilbert Function Bounded by SOS Degree for Max-CUT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Toric Varieties × SOS Degree for Max-CUT
- **Recorded**: 2026-05-10 20:44 UTC
- **Entry ID**: `5eaf43fc827e`

Statement

For every 3-CNF formula Φ on n variables, the Hilbert function H_I_Φ(d) of its associated toric ideal I_Φ satisfies H_I_Φ(d) ≤ O((n)^{d/2}) for all d ≤ SOS_degree(Φ).

Rationale

Toric varieties encode combinatorial structures of monomial ideals, and their Hilbert functions reflect the complexity of the ideal's generators. Linking this to SOS degree could expose hidden geometric constraints on semidefinite relaxations of Max-CUT.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6901ad7c.py", line 82, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_6901ad7c.py", line 56, in run_trial
    monomial_dict[abs(var)][var] += 1
    ~~~~~~~~~~~~~~~~~~~~~~~^^^^^
KeyError: -29
```

Judge reasoning

Test crashed with KeyError before producing results, preventing evaluation of conjecture validity. | next: Debug and fix the KeyError in test_6901ad7c.py, then re-run trials

---

Polymatroid Rank Lower Bound for Monotone k-CLIQUE

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Polymatroid Theory × Monotone Circuit Size for k-CLIQUE
- **Recorded**: 2026-05-10 21:20 UTC
- **Entry ID**: `aae5f38e2e54`

Statement

For a k-CLIQUE instance with n vertices, the polymatroid rank of its incidence matrix is Ω(n) if the instance is a complete graph, and O(log n) for any instance with a monotone circuit of size poly(n).

Rationale

Polymatroid rank captures the interdependency of clauses, which mirrors the combinatorial complexity of monotone circuits. High rank implies structural constraints that may necessitate exponential circuit size.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_62dc92df.py", line 100, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_62dc92df.py", line 68, in run_trial
    rank = rank_polymatroid(clauses)
           ^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_62dc92df.py", line 49, in rank_polymatr
    gaussian_elimination(matrix)
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_62dc92df.py", line 29, in gaussian_elim
```

```
    factor = Fraction(1, matrix[i][i])
             ^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/fractions.py", line 281, in __new__
    raise ZeroDivisionError('Fraction(%s, 0)' % numerator)
ZeroDivisionError: Fraction(1, 0)
```

Judge reasoning

Test crashed with ZeroDivisionError during Gaussian elimination, preventing result collection. The crash suggests potential issues with matrix construction or singular matrices, but no conclusive data was produced. | next: Debug the Gaussian elimination implementation to handle singular matrices or zero division errors, then re-run tests with controlled examples.

Coxeter Polynomial Root Count Inverse Proportional to Resolution Length in 3-SAT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Coxeter Group Representation Theory × Resolution Proof Length
- **Recorded**: 2026-05-10 21:50 UTC
- **Entry ID**: `102ae91c53d3`

Statement

For a 3-SAT instance φ, let C_φ be its associated Coxeter group element derived via clause-to-reflection mapping. The number of distinct roots of the Coxeter polynomial Π_φ(x) satisfies |roots(Π_φ)| ≥ $\log_2$(ρ(φ)) + 1, where ρ(φ) is the resolution proof length of φ.

Rationale

Coxeter polynomials encode symmetry invariants of reflection groups, which may mirror structural constraints in clause interaction graphs. These invariants could expose hidden algebraic regularities that govern proof complexity trade-offs.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7e3581c5.py", line 152, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7e3581c5.py", line 120, in run_trial
    C += clause_to_reflection(clause)
         ^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7e3581c5.py", line 77, in clause_to_ref
    reflection[-var-1][-var-1] = Fraction(-2)
    ~~~~~~~~~~^^^^^^^^
IndexError: list index out of range
```

Judge reasoning

Test crashed during clause-to-reflection mapping, preventing evaluation of conjecture validity | next: Fix the IndexError in clause_to_reflection by verifying reflection matrix dimensions

---

Spectral Concentration of Moment Matrices in Max-CUT SOS Hierarchy

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Random Matrix Theory × SOS Degree for Max-CUT
- **Recorded**: 2026-05-10 21:59 UTC
- **Entry ID**: `5167cdc13e84`

Statement

For a random Max-CUT instance with n variables, the spectral norm of the degree-d moment matrix M_d is at least Ω(√n) if d < c log n for some constant c. Thus, any SOS proof system achieving a 0.879-approximation for Max-CUT must have d ≥ Ω(log n).

Rationale

The spectral norm of the moment matrix reflects the ability to capture the structure of the Max-CUT problem. If the norm is too small, the SOS hierarchy cannot approximate the problem well, implying a lower bound on the required degree.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_12e5d29b.py", line 89, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_12e5d29b.py", line 63, in run_trial
    M_d = moment_matrix(instance, d)
          ^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_12e5d29b.py", line 55, in moment_matrix
    new_M[i+1][j+1] += instance[i][l] * M[j+1][k-1+l]
                                        ~~~~~~^^^^^^^
IndexError: list index out of range
```

Judge reasoning

Test crashed with IndexError before collecting data; cannot assess conjecture validity | next: Debug the moment_matrix implementation to fix index out-of-range error

---

Noncommutative Fourier Coefficient Gap in Read-Twice BPs for IP_2

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Noncommutative Harmonic Analysis × Read-Twice Branching Programs

- **Recorded**: 2026-05-10 22:09 UTC
- **Entry ID**: a310eda906ef

Statement

For any read-twice BP P computing the inner product mod 2 function IP_2, the maximum absolute value of its noncommutative Fourier coefficients is Ω(n), whereas for read-once BPs, it is O(log n).

Rationale

Noncommutative harmonic analysis can capture the non-abelian structure of BP transitions, which might exhibit different Fourier coefficient growth rates between read-once and read-twice models.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 0.32s

```
oncommutative_fourier_coefficient', 'metric_value': 0.9259259259259259, 'instances_tested': 1, 'con
TRIAL: {'metric_name': 'noncommutative_fourier_coefficient', 'metric_value': 1.0, 'instances_tested
TRIAL: {'metric_name': 'noncommutative_fourier_coefficient', 'metric_value': 0.9285714285714286, 'in
TRIAL: {'metric_name': 'noncommutative_fourier_coefficient', 'metric_value': 1.0, 'instances_tested
TRIAL: {'metric_name': 'noncommutative_fourier_coefficient', 'metric_value': 0.9629629629629629, 'in
TRIAL: {'metric_name': 'noncommutative_fourier_coefficient', 'metric_value': 0.9615384615384616, 'in
TRIAL: {'metric_name': 'noncommutative_fourier_coefficient', 'metric_value': 1.0, 'instances_tested
TRIAL: {'metric_name': 'noncommutative_fourier_coefficient', 'metric_value': 1.0833333333333333, 'in
TRIAL: {'metric_name': 'noncommutative_fourier_coefficient', 'metric_value': 0.896551724137931, 'ins
TRIAL: {'metric_name': 'noncommutative_fourier_coefficient', 'metric_value': 1.0769230769230769, 'in
TRIAL: {'metric_name': 'noncommutative_fourier_coefficient', 'metric_value': 1.0, 'instances_tested
TRIAL: {'metric_name': 'noncommutative_fourier_coefficient', 'metric_value': 1.1153846153846154, 'in
RESULT: FALSIFIED counterexample="" first_failing_seed=11
```

Judge reasoning

All tested instances have n=1 (implied by instances_tested=1), preventing observation of Ω(n) growth. The critic challenges the validity of the test setup. | next: Test with larger n (e.g., n=100) to observe scaling behavior of the metric

---

Additive Energy Inverse Proportional to Disjointness Discrepancy

- **Verdict**: INCONCLUSIVE
- **Bridge**: Additive Combinatorics × Communication Complexity of Disjointness
- **Recorded**: 2026-05-10 22:42 UTC
- **Entry ID**: 4ab417bc1c0b

Statement

For a disjointness function f: {0,1}^n × {0,1}^n → {0,1}, the additive energy E(f) of its communication matrix satisfies E(f) = Θ(1/disc(f)), where disc(f) is the discrepancy of f under uniform distributions.

Rationale

Additive energy captures the number of additive configurations in the communication matrix, which directly influences the discrepancy via the structure of overlapping supports. High additive energy implies structured overlaps, reducing discrepancy.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_44b6282a.py", line 73, in <module>
    seeds = [int(x) for x in input().split()] if len(sys.argv) > 1 else list(range(2, 30))  # Default to
                             ^^^^^^^
EOFError: EOF when reading a line
```

Judge reasoning

Test crashed before producing data, preventing evaluation of support fraction or counterexamples | next: Fix the test to handle input properly and re-run with sufficient seed coverage

---

Moment-Matrix Spectral Norm Inverse Proportional to Communication Complexity Lower Bound for Max-CUT

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Real Algebraic Geometry (Moment Matrices, Positivstellensatz) × Communication Complexity of Max-CUT Instances
- **Recorded**: 2026-05-10 23:07 UTC
- **Entry ID**: `9a3f01389b5d`

Statement

For a Max-CUT instance on n variables, the spectral norm of its degree-d moment matrix is inversely proportional to the communication complexity lower bound, i.e., ||M_d|| ≥ c * (log(1/ε)) / (log n), where ε is the approximation ratio and c is a constant.

Rationale

The spectral norm of the moment matrix reflects the structure of the polynomial constraints in the SOS hierarchy. By linking it to communication complexity, we can uncover how the algebraic structure of the problem influences its communication requirements, potentially revealing new lower bound techniques.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.13s

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dd896e7d.py", line 70, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dd896e7d.py", line 50, in run_trial
    norm = spectral_norm(M_d)
           ^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_dd896e7d.py", line 31, in spectral_norm
    v = [Fraction(1, math.sqrt(n)) for _ in range(n)]
         ^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/fractions.py", line 277, in __new__
    raise TypeError("both arguments should be "
TypeError: both arguments should be Rational instances
```

Judge reasoning

Test crashed due to invalid Fraction arguments; no data collected to assess conjecture | next: Fix the Fraction initialization error by using integer denominators

---

Schur-Weyl Decomposition Entropy Gap for Permanent vs Determinant

- **Verdict**: INCONCLUSIVE
- **Bridge**: Schur-Weyl Duality × Boolean Circuit Size for Permanent vs Determinant
- **Recorded**: 2026-05-10 23:56 UTC
- **Entry ID**: 01f932c44333

Statement

For all $n \geq 2$, the entropy of the Schur-Weyl decomposition of the permanent polynomial (measured in bits) exceeds that of the determinant polynomial by $\Omega(n \log n)$, where entropy is defined as $-\Sigma p_\lambda \log p_\lambda$ for the distribution over irreducible components λ.

Rationale

Schur-Weyl duality decomposes tensor powers into symmetric/antisymmetric parts, revealing structural differences between permanent and determinant. The entropy gap could reflect inherent complexity differences, as higher entropy implies more complex decomposition patterns.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 124, elapsed: 240.00s

```
TIMEOUT after 240s
```

Judge reasoning

Test timed out without producing data; critic challenges validity of results | next: Run test with extended timeout and verify implementation correctness